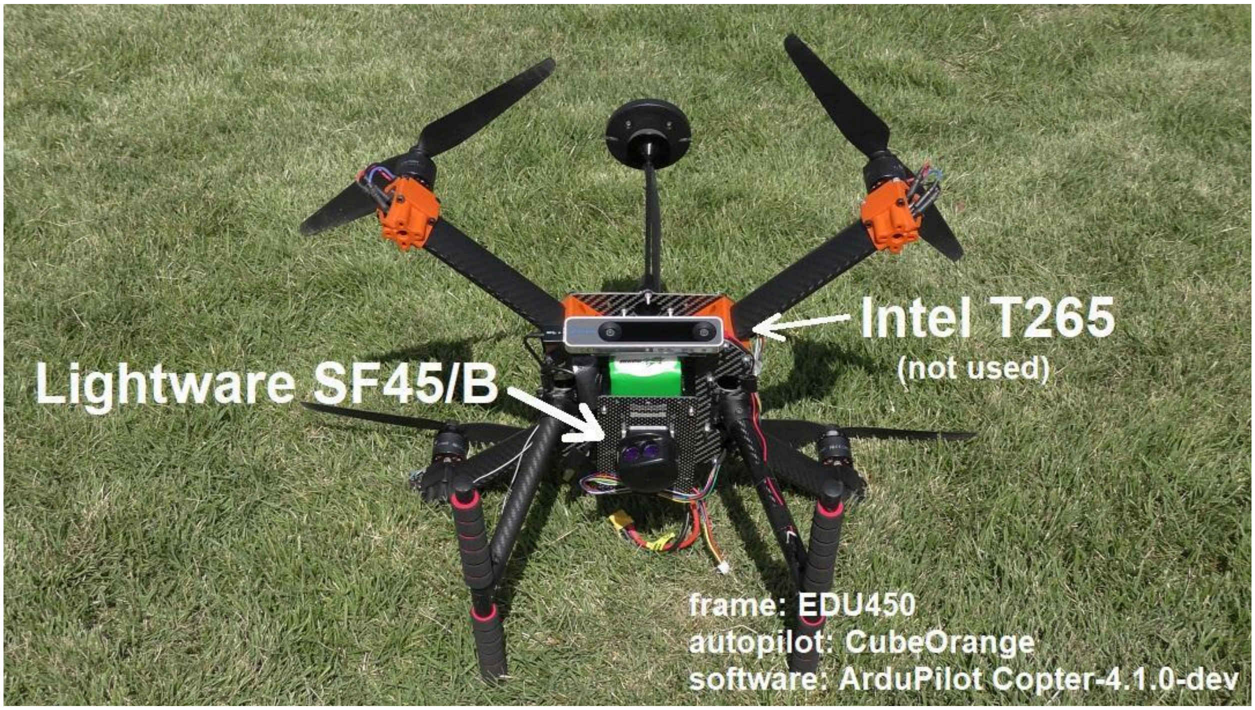
The SF45/B microLiDAR is a very small **2D scanning LiDAR sensor** made by LightWare LiDAR for:

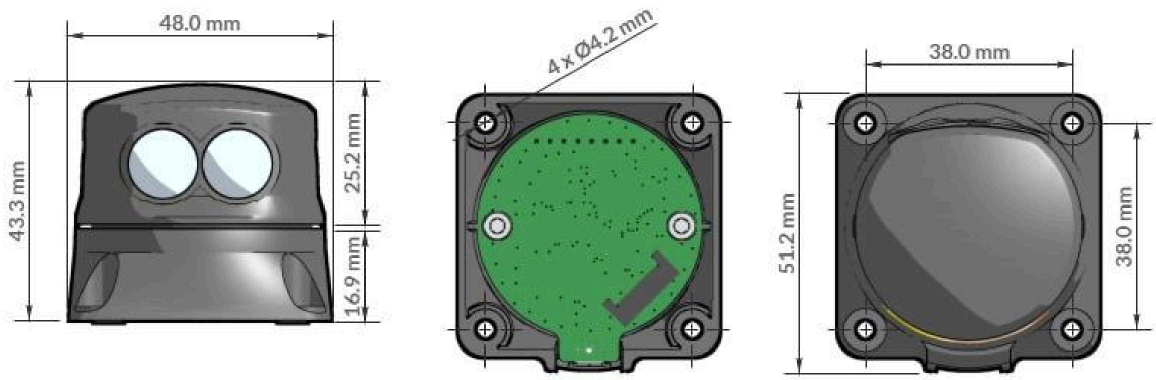
- Robotics
- Drones (UAV)
- SLAM mapping
- Obstacle avoidance
- Autonomous navigation


SF45
Class 1M
Laser
S45-
15197


Intel T265
(not used)
Lightware SF45/B
frame: EDU450
autopilot: CubeOrange
software: ArduPilot Copter-4.1.0-dev


1
SF45
Rev3


E320444



Main Specifications

Feature	Value
Scan Angle	Up to 320°
Range	0.2m to 50m
Weight	~58–59g
Scan Speed	Up to 5000 readings/sec
Interfaces	USB, UART, I2C
Laser Type	905nm Class 1M
Use	Indoor + Outdoor

([DigiKey])

Works With

- Raspberry Pi
- Arduino
- Espressif Systems ESP32
- PX4
- ArduPilot
- ROS robotics systems

([LightWare LiDAR])

What You Can Build

1. Autonomous robot
2. Drone obstacle avoidance
3. 2D room mapping
4. SLAM navigation
5. AI security rover
6. Warehouse robot
7. Smart terrain scanner

Connection Example

SF45/B → Raspberry Pi 5

- TX → RX
- RX → TX
- GND → GND
- 5V → 5V USB/UART

You can read LiDAR data using:

- Python
- ROS
- C++
- Arduino IDE

Example Python Use

```
import serial

ser = serial.Serial('/dev/ttyUSB0', 115200)

while True:
    data = ser.readline()
    print(data)
```

Official Resources

- [LightWare SF45/B Official Page]
- [SF45/B Product Guide PDF]
- [DigiKey SF45/B Info]

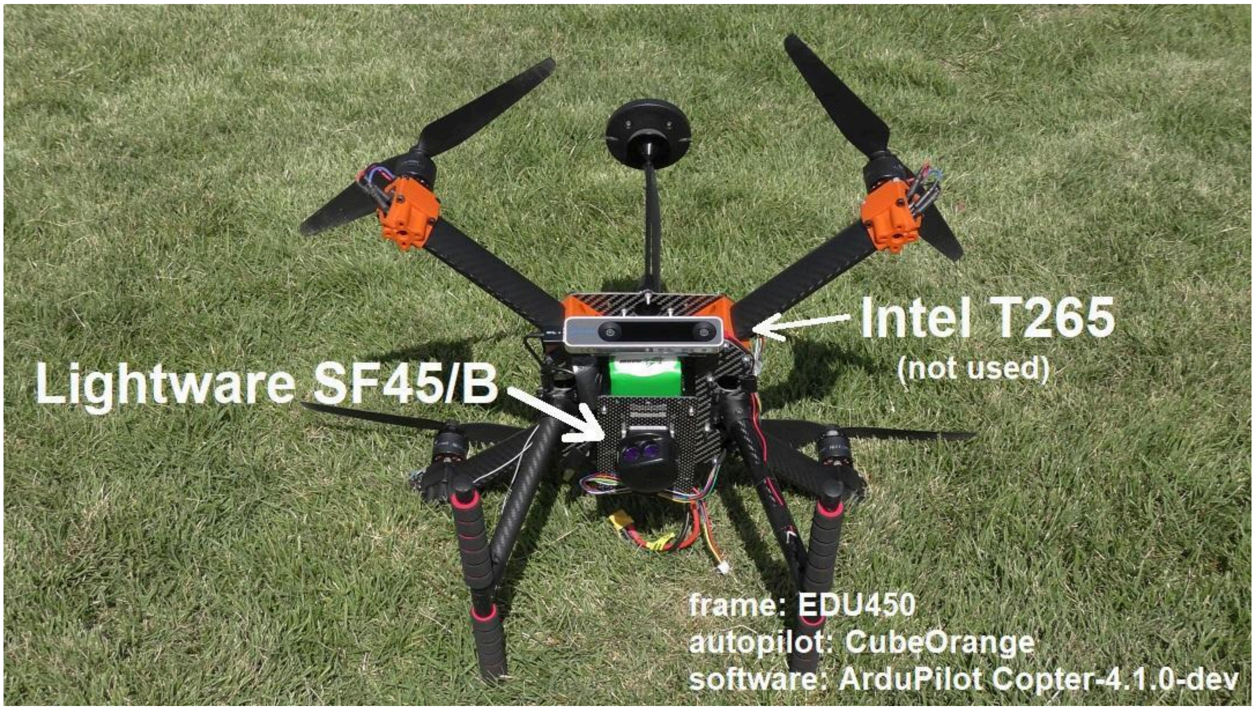
Price is usually around:

- $449 USD

- ₹45,000–₹50,000 in India

(DigiKey)

Advanced SF45/B microLiDAR projects 🔥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LSROBOT
THE WORLD'S LEADING 3D SLAM
AMR&UNMANNED FORKLIFT


Raspberry Pi 4 Model B
©Raspberry Pi 2018

The SF45/B microLiDAR is powerful for:

- SLAM
- Autonomous robots
- AI navigation
- Drone safety
- 3D mapping
- Defense robotics

It supports:

- Raspberry Pi
- ROS
- ArduPilot
- PX4
- Arduino
- ESP32 (LightWare LiDAR)

Advanced Project Ideas

1. Autonomous AI Rover

Build a robot that:

- Creates room maps
- Avoids obstacles
- Navigates automatically
- Uses AI object detection

Components

- Raspberry Pi 5
- SF45/B
- Motor driver
- Camera
- Battery
- ROS2

Skills

- SLAM
- ROS
- Python
- OpenCV

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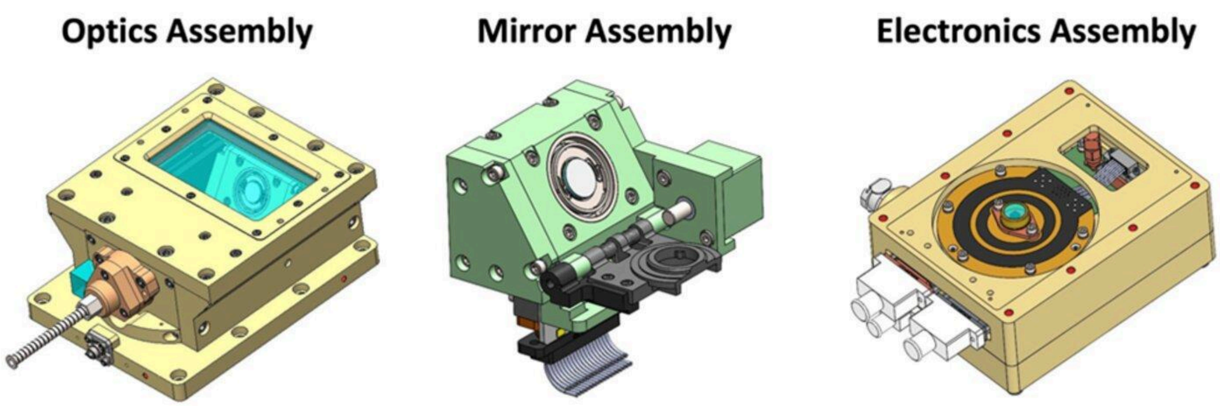
2. Defense Surveillance Rover


TRITON
ROBOTIC



Features:

- Night patrol

- Human detection
- Thermal camera
- Silent obstacle avoidance
- Remote control + autonomous mode

Add:

- Thermal camera
- GPS
- LoRa communication
- AI target tracking

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3. Drone Obstacle Avoidance System

Use SF45/B on drone:

- Tree avoidance
- Building detection
- Terrain following
- Autonomous landing

Works with:

- PX4
- ArduPilot (Ncnynl Docs)

Advanced version:

- AI route planning
- Swarm drones
- Indoor GPS-free navigation

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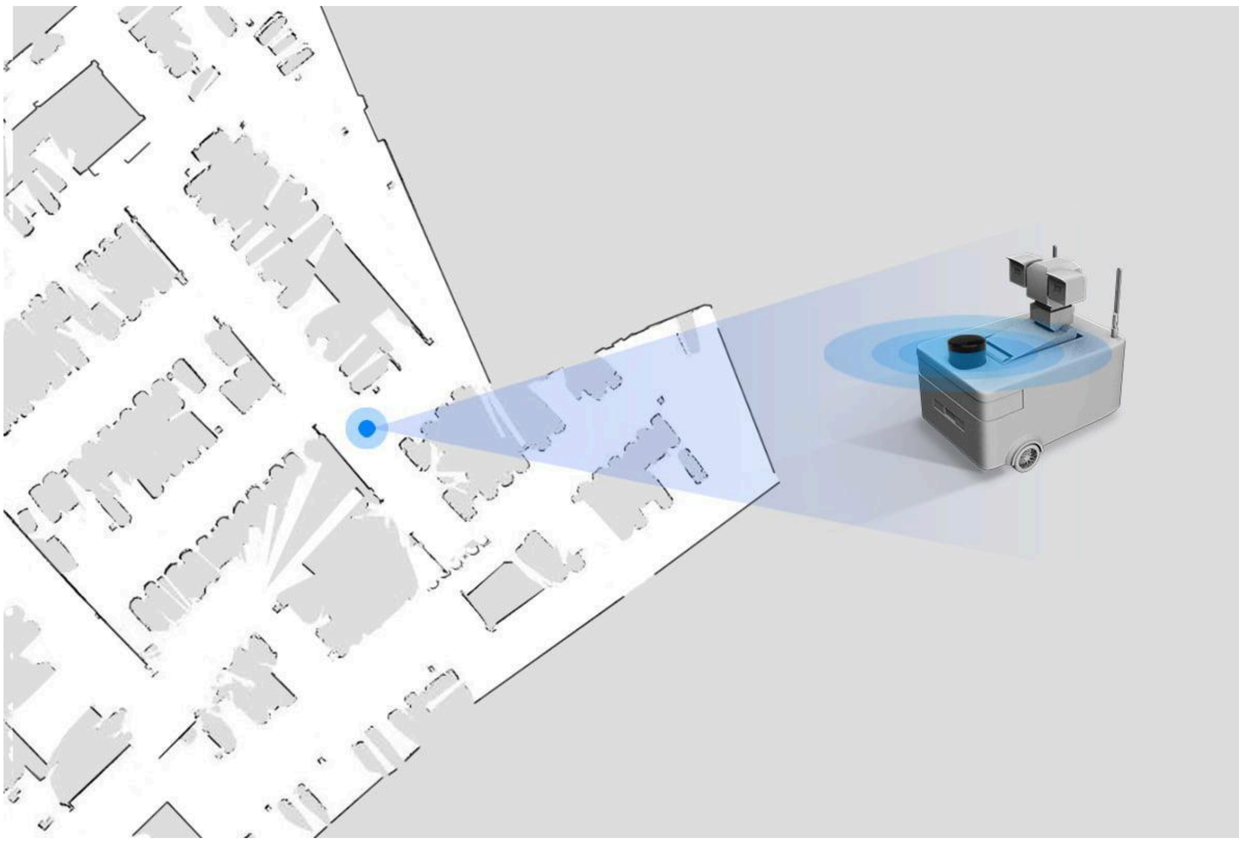
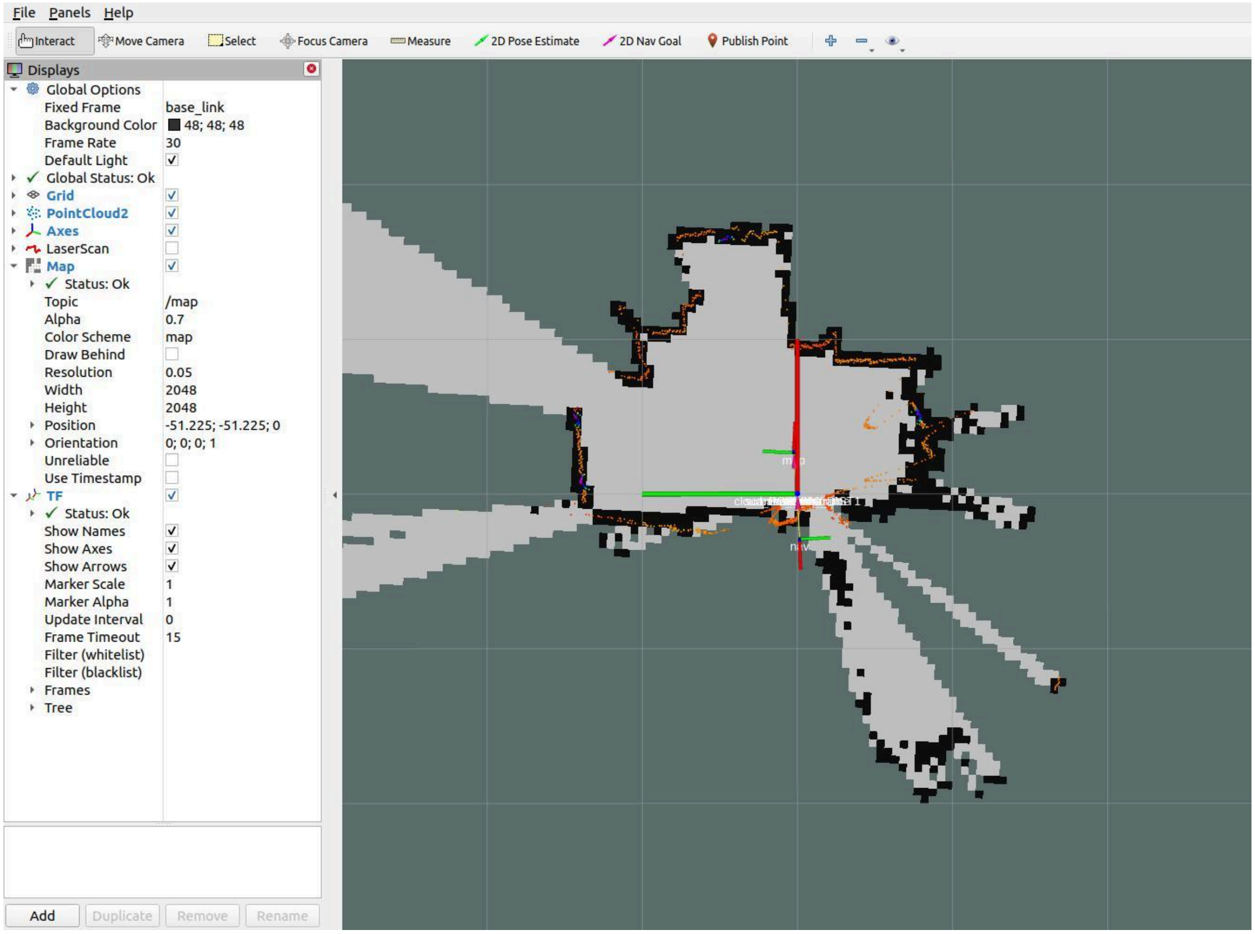
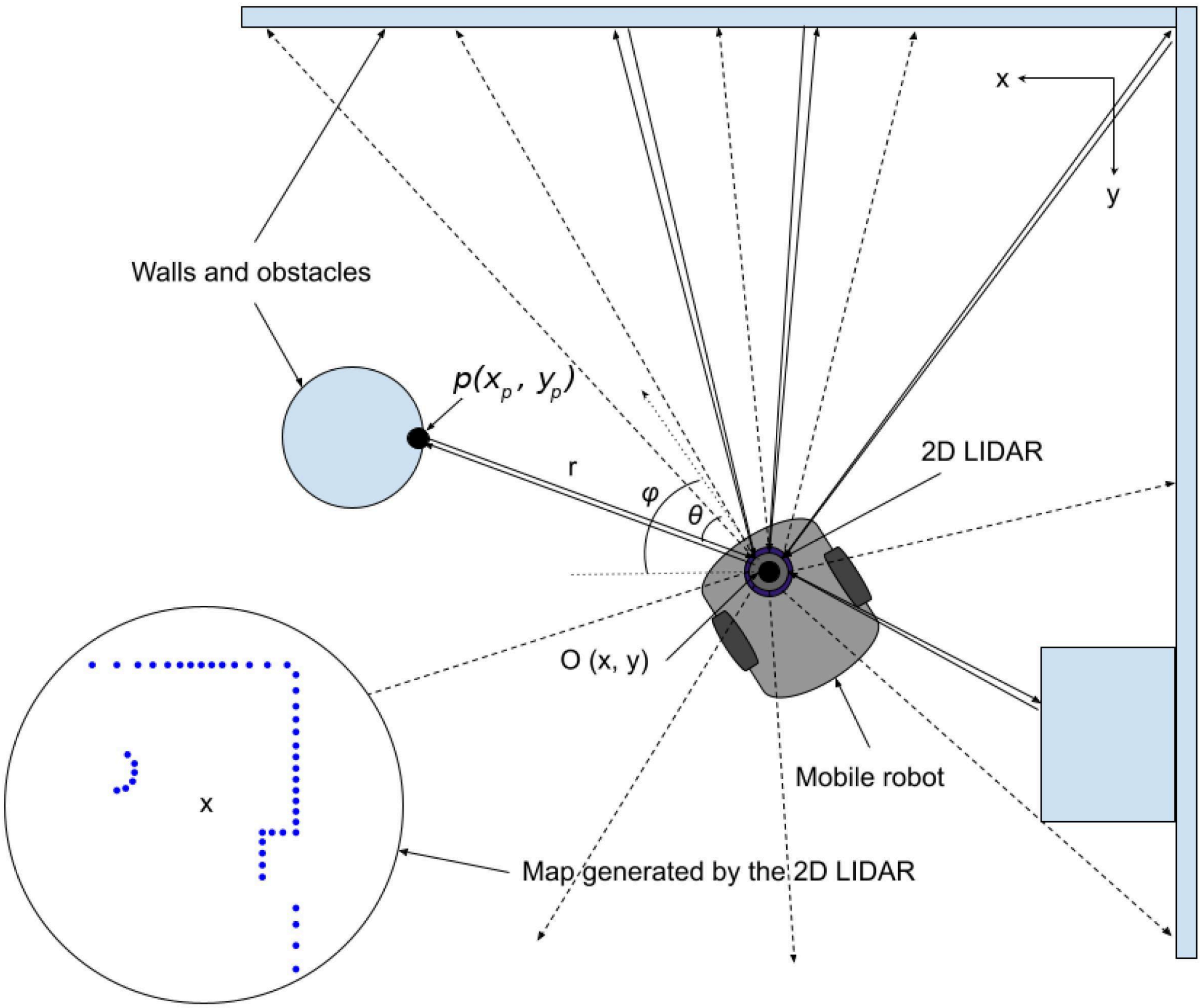
4. 2D/3D SLAM Mapping Robot

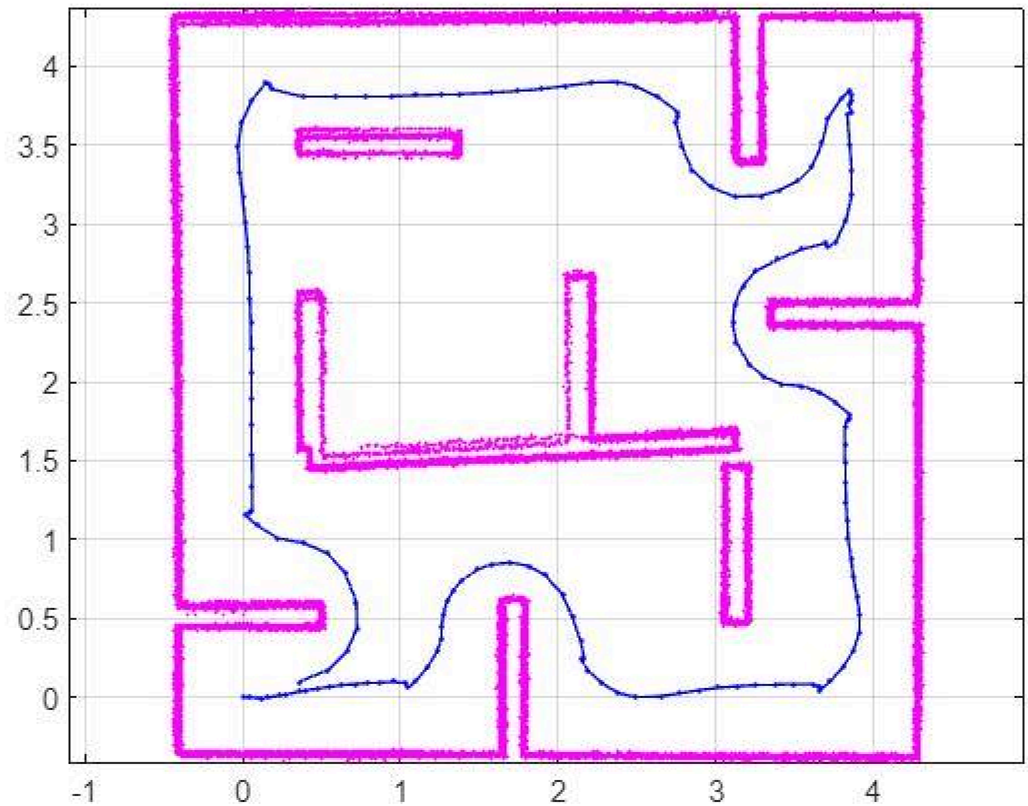
(**a**) (**b**)



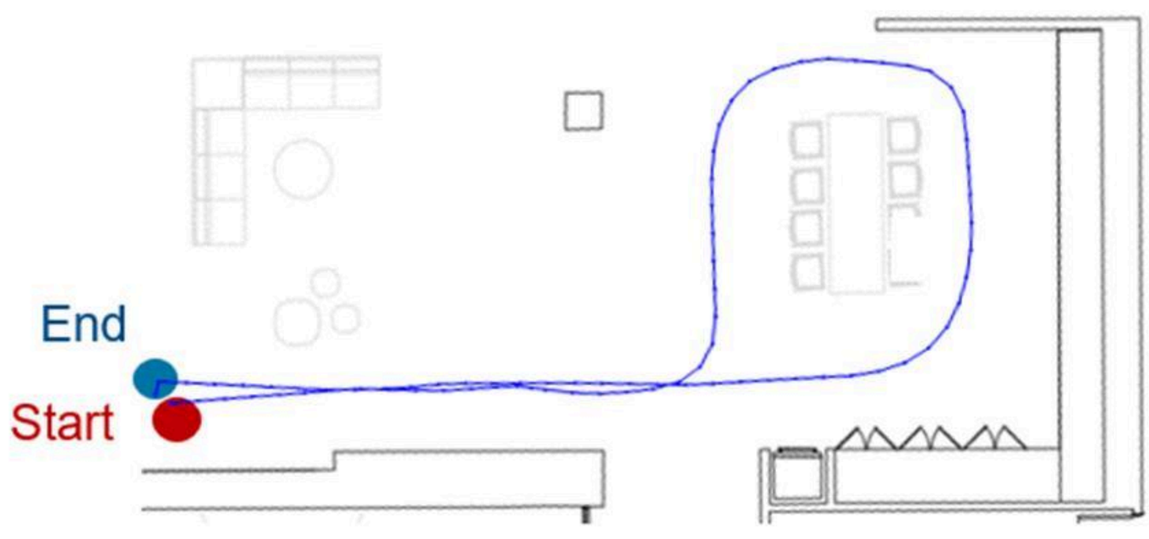

File
Panels
Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Nav Goal
Publish Point
Displays
Global Options
Fixed Frame
base_link
Background Color
48; 48; 48
Frame Rate
30
Default Light
Global Status: Ok
Grid
PointCloud2
Axes
LaserScan
Map
Status: Ok
Topic
/map
Alpha
0.7
Color Scheme
map
Draw Behind
Resolution
0.05
Width
2048
Height
2048
Position
-51.225; -51.225; 0
Orientation
0; 0; 0; 1
Unreliable
Use Timestamp
TF
Status: Ok
Show Names
Show Axes
Show Arrows
Marker Scale
1
Marker Alpha
1
Update Interval
0
Frame Timeout
15
Filter (whitelist)
Filter (blacklist)
Frames
Tree
Add
Duplicate
Remove
Rename



x
y
Walls and obstacles
p(x_p , y_p)
r
φ
θ
2D LIDAR
O (x, y)
Mobile robot
x
Map generated by the 2D LIDAR



4
3.5
3
2.5
2
1.5
1
0.5
0
-1
0
1
2
3
4



End
Start

Robot scans:

- Rooms
- Warehouses
- Tunnels
- Factories

Software:

- ROS2
- Cartographer
- RViz
- Python

Output:

- Live map generation
- Navigation path planning

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5. AI Smart CCTV Turret

Features:

- 360° scanning
- Intruder detection
- Auto tracking
- Distance measurement
- Face recognition

Use:

- SF45/B
- ESP32
- Raspberry Pi
- Servo motors
- OpenCV AI

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6. Autonomous Farming Robot

Tasks:

- Crop row navigation
- Obstacle detection
- Weed identification
- Smart spraying

Can combine:

- YOLO AI
- GPS RTK
- Thermal imaging

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7. Indoor Delivery Robot

Like mini warehouse robots:

- Self navigation
- Path planning
- Shelf detection
- Human avoidance

Technologies:

- ROS2
- SLAM
- LiDAR localization

Ultra Advanced Ideas

AI + LiDAR Fusion

Combine:

- Camera vision
- Thermal vision
- LiDAR depth
- AI detection

Used in:

- Defense
- Search & rescue
- Smart factories
- Autonomous vehicles

Multi-Sensor Edge AI System

Use:

- Raspberry Pi 5
- NVIDIA Jetson
- SF45/B
- mmWave radar
- Thermal camera

Run:

- YOLOv8
- TensorRT
- Edge AI

Best Software Stack

Purpose	Software
Robotics	ROS2
AI	PyTorch / YOLO
Vision	OpenCV
Mapping	Cartographer
Drone	PX4 / ArduPilot
Edge AI	TensorRT
Programming	Python + C++

Skills You Will Master

- Embedded systems
- Robotics
- SLAM
- Computer vision
- AI/ML
- Sensor fusion
- Autonomous systems
- Linux
- ROS2

Official resources:

- LightWare SF45/B Official
- PX4 SF45/B Setup Guide

The SF45/B can scan up to 320° and detect obstacles up to 50 meters while weighing only about 58–59g. (DigiKey)